

MAKSYM SYROMOLOT

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SUMMARY

Full-stack software engineer with production experience building Python and TypeScript/Node.js backend services, RESTful APIs, and React frontends. Background spans LLM-powered application development, QA automation, and cloud-deployed services on Kubernetes. Hands-on with AI tool integration (structured outputs, tool calling, agent workflows), automated testing (Playwright, Pytest), and CI/CD pipelines from government software delivery. Available full-time in Toronto; Seneca Polytechnic Advanced Diploma, GPA 4.0.

EDUCATION

Computer Programming & Analysis, Advanced Diploma

Seneca Polytechnic

GPA: 4.0 / 4.0

Sept 2023 – May 2026

WORK EXPERIENCE

Associate Software Engineer | OpenPolicy

May 2025 – Dec 2025

- Built and operated full-stack production services (Python & TypeScript/Node.js), owning end-to-end feature delivery from data model design through deployment and monitoring in a Kubernetes-based cloud environment.
- Developed AI-powered document scanning and bill interpretation systems; integrated third-party APIs with structured data contracts, JWT-based security, and Docker-based deployments.
- Designed and maintained RESTful APIs with API-first approach; enforced security standards and wrote clear API documentation for cross-functional consumers.
- Collaborated in a Git-based Agile environment; participated in code reviews, managed tickets, and maintained technical documentation for APIs and system components.

QA Automation Tester | Ministry of Education, Govt. of Ontario

Sept 2024 – Apr 2025

- Developed and documented test plans and test cases for critical government regulatory software, covering business-rule and edge-case scenarios.
- Built and maintained automated E2E test pipelines with Playwright, integrated into CI/CD (Jenkins, GitHub Actions), reducing defect escape rate and improving release confidence.
- Designed validation suites covering REST API payloads (JSON/XML), async workflows, and data-driven scenarios; logged defects with reproducible steps and impact assessment.

Writer, Programmer & Organizer | Media Bober (NGO) & Student Union Priama Diia

Jun 2024 – Present

- Authored technical guides on Linux, open-source tooling, and security practices; wrote a full-stack website (Python & TypeScript) for the student union.
- Organized digital literacy workshops, fostering technical knowledge-sharing across a diverse community.

TECHNICAL PROJECTS

Job Hunter Agent — Python, Pydantic, OpenRouter, multi-stage validation

- Built an LLM-powered job intelligence pipeline that ingests PDFs and a resume to produce gap reports, fit scores, and tailored cover letters using function calling and structured outputs.
- Implemented an autonomous research agent with a multi-turn tool-calling loop; engineered Pydantic schemas injected into LLM prompts for structured extraction. Provider-agnostic backend supports OpenRouter, Anthropic, Ollama, and vLLM.

Hotel Reservation System — Java, JavaFX, JPA/Hibernate, MySQL, 3-tier MVC

- Architected a full-stack desktop hotel system using JavaFX with 3-tier MVC, JPA/Hibernate ORM, and MySQL; implemented 5 design patterns (Strategy, Observer, Factory, Decorator, Singleton).

- Built role-based admin module, self-service booking kiosk with dynamic pricing, and a real-time waitlist system using the Observer pattern.

Scriptorium — *Node.js, Express, React, MongoDB, OpenRouter API, Docker, GitHub Actions*

- Architected a decoupled full-stack system; built RESTful APIs for authentication, media tracking, and gamification with JWT security.
- Integrated OpenRouter API for AI-driven content recommendations using structured JSON outputs and tool calling; containerized with Docker and maintained via GitHub Actions CI.

DangerAlert — *MySQL, relational schema design, SQL*

- Led a team of 3 to design a normalized relational schema from business requirements; wrote and validated complex SQL queries (joins, aggregations, subqueries) to produce operational reports and catch data anomalies.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, Bash, SQL; learning Go

AI & LLM: Tool calling, structured outputs, agent workflows; OpenRouter, Anthropic, Ollama APIs; Pydantic schema engineering

Frontend: React, Next.js, JavaFX; component-driven UI development

Backend & APIs: Node.js, Express, RESTful API design, JWT/OAuth 2.0, Spring Framework; Java backend services

QA & Testing: Test plan/case design; Playwright (E2E), Pytest, JUnit; functional, regression, integration, data-driven testing; defect logging

Databases: PostgreSQL, MySQL, MongoDB, BigQuery; schema design & normalization; JPA/Hibernate ORM

Cloud & DevOps: Docker, Kubernetes, AWS, OpenShift, GitHub Actions, Jenkins, CI/CD, Linux

Agile & Tools: Scrum/Kanban, JIRA-style workflows, code review, Git, GitHub Copilot, technical documentation